Topic:

Q. What is echelon form?

Echelon form:

- A matrix is in echelon form (or row echelon form) if it has the following three properties:

i) All non-zero rows are above any row of all zeroes.

ii) The number of zero preceding the first non-zero element in a row is less than that the number of such zeros in the succeeding (or next row.

iii) The first non-zero element in every row is unity.

⇒ When a matrix is converted in echelon form, then the number of non-zero rows of the matrix is known as the rank of the matrix A

example:

matrix $A = \begin{bmatrix} 1 & 3 & -2 & 6 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in the echelon matrix.

Q. What is pivot element or row leader?

⇒ In echelon matrix, the first non-zero entry of a row is called pivot element or row leader.

e.g.

pivot element or row leader

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

What is pivot Column?

Pivot Column:

The Column which has the pivot element is called pivot Column.

What is reduced echelon form?

Reduced echelon form:

⇒ A matrix is in reduced echelon form if it has the following five properties:

i) All non-zero rows are above any row of all zeroes.

⇒ The number of zero preceding the first non-zero element in a row is less than that the number of such zeros in the succeeding (or next row).

⇒ First non zero number in a row should be one.

⇒ Leading element in any row must be the only non-zero element in that column.

Example:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: Every matrix in reduced echelon form is also in echelon form but the converse is not true.

• **Examples of reduced row echelon form:**

Q#19

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Elementary row Operation:

The following operations could only be performed:

i) Interchange of any two rows

$R_{ij} \to R_i \longleftrightarrow R_j \quad i \neq j$

ii) multiplying of a row of A by any non-zero real or complex number.

$R_i \to \lambda R_i , \quad \lambda \neq 0$

iii) Addition of a scalar multiple of one row, row to another.

$R_i \to R_i + \lambda R_j , \quad i \neq j$

Row equivalence:

Let A and B be two matrices. A is row equivalent to B $(A \sim B)$ iff there is a finite sequence of e.r.o that transform A into B.

$(A \sim B) \; (B \sim A)$

Uniqueness of ref:

⇒ each matrix is row equivalent to one and only reduced echelon matrix.

Convert in reduce echelon form.

$$\begin{bmatrix} -1 & 1 & -1 & -2 \\ 2 & 0 & 0 & 3 \\ -1 & 0 & -1 & -2 \\ -3 & 0 & 2 & -1 \end{bmatrix}$$

$R_1 \leftarrow (-1)R_1$
$$\begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & 0 & 0 & 3 \\ -1 & 0 & -1 & -2 \\ -3 & 0 & 2 & -1 \end{bmatrix}$$

$R_2 \leftarrow R_2 + (-2)R_1$

$R_3 \leftarrow R_3 + R_1$

$R_4 \leftarrow R_4 + 3R_1$
$$\begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & -3 & 5 & 5 \end{bmatrix}$$

$\sim R_2 \leftarrow R_2 + (-2)R_1$
$$\begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & -2 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & -3 & 5 & 5 \end{bmatrix}$$

$\cancel{R_4 \leftarrow R_4 + (-3)R_3}$

$\sim R_2 \leftarrow R_2 + R_3$
$$\begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & -3 & 5 & 5 \end{bmatrix}$$

$R_1 \leftarrow R_1 + R_2$
$R_3 \leftarrow R_3 + R_2$
$R_4 \leftarrow R_4 + 3R_2$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$R_3 \leftarrow R_3 - 3R_4$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$R_1 \leftarrow R_1 + R_3$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$R_2 \leftarrow R_2 + 2R_3$
$R_4 \leftarrow R_4 + R_3$

$$\begin{bmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & -15 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & -5 \end{bmatrix}$$

$R_4 \leftarrow -\frac{1}{5} R_4$

$$\begin{bmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & -15 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_1 \leftarrow R_1 + 6R_4$
$R_2 \leftarrow R_2 + 15R_4$
$R_3 \leftarrow R_3 + 7R_4$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Hints to Solve:

- Try to make 1st element of 1st row is 1

$$\begin{bmatrix} 1 & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

- Try to make element 0 on 1st column then 2nd column.

$$\begin{bmatrix} 1 & - - & - - \\ 0 & - - & - - \\ 0 & - - & - - \end{bmatrix}$$

Try to make diagonal element 1

$$\begin{bmatrix} 1 & - & - \\ - & 1 & - \\ - & - & 1 \end{bmatrix}$$

Gauss Elimination Method:

=> it is used to Solve system of linear equation.
=> it is also called the forward elimination method.
=> it involves two steps.
=> Convert the given matrix into augmented matrix.
=> Apply row operation for obtain the echelon form.
=> Use back substitution method for Solving or finding the values of variab.

Example:

free variables.

basic variable

$$2x - 7y - 10z = -17$$
$$5x + y + 3z = 14$$
$$x + 10y + 9 = 7$$

Given System

$$2x - 7y - 10z = -17 \quad \text{--- (i)}$$
$$5x + y + 3z = 14 \quad \text{--- (ii)}$$
$$x + 10y + 9 = 7 \quad \text{--- (iii)}$$

$$A = \left[\begin{array}{ccc|c} 2 & -7 & -10 & -17 \\ 5 & 1 & 3 & 14 \\ 1 & 10 & 9 & 7 \end{array}\right]$$ Augmented matrix

Q. **Example 2:**

$$x - y + 2z = 3$$
$$x + 2y + 3z = 5$$
$$3x - 4y - 5z = 13$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 2 & 3 \\ 3 & -4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 13 \end{bmatrix}$$

$$A \quad X = B$$

$$(A'-B) = \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 1 & 2 & 3 & 5 \\ 3 & -4 & -5 & 13 \end{array}\right]$$

$\sim R_2 \leftarrow R_2 - R_1$

$\sim R_3 \leftarrow R_3 - 3R_1$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & -7 & -11 & -22 \end{array}\right]$$

$\sim R_3 \leftarrow 3R_3 + (7)R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & -32 & 69 \end{array}\right]$$

$-32z = 69 \Rightarrow \boxed{z = 2}$

$3y + z = 2$

$\cancel{3y + 2 = 2} \Rightarrow \boxed{y = 0}$

$x - y + 2z = 3$

$x - 0 + 2(2) = 3$

$\boxed{x = -1}$

Verification:

$x - y + 2z = 3$

$-1 - 0 + 2(2) = 3$

$-1 + 4 = 3$

$\boxed{3 = 3}$

Example 3

Solve the System Using:
(i) Gauss elimination
(ii) Gauss Jordan method.

(1) $2x + y + z = 10$
$3x + 2y + 3z = 18$
$x + 4y + 9z = 16$

$$\begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{matrix} 10 \\ 18 \\ 16 \end{matrix}$$

$A \; X = B$

$$(A : B) = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array}\right]$$ Augmented matrix

$R_1 \leftrightarrow R_2$
$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$
$R_3 \leftarrow R_3 - 3R_1$

$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -24 & -30 \end{array}\right]$$

$R_3 \to 7R_3 - 10R_2$

$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & 0 & 2 & 10 \end{array}\right]$$

$$x + 4y + 9z = 16$$
$$-7y - 17z = -22$$
$$2z = 10$$

$$\boxed{z = 5}$$

$$-7y - 17(5) = -22$$
$$-7y = 63$$
$$\boxed{y = -9}$$

$$x + 4(-9) + 9(5) = 16$$
$$x - 36 + 45 = 16$$
$$\boxed{x = 7}$$

Verification:

$$2(7) + (-9) + 5 = 10$$
$$14 - 9 + 5 = 10$$
$$19 - 9 = 10$$
$$10 = 10$$

Gauss jordan elimination. (Q. imp)

Steps:

1) Form augmented matrix
2) Apply row operation to form reduce echelon form.

Example:

$$2x+y+z=10$$
$$3x+2y+3z=18$$
$$x+4y+9z=16$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$AX=B$$

Augmented matrix:-

$$(A:B) = \begin{bmatrix} 2 & 1 & 1 & : & 10 \\ 3 & 2 & 3 & : & 18 \\ 1 & 4 & 9 & : & 16 \end{bmatrix}$$

$R_1 \longleftrightarrow R_3$

$R_3 \longleftrightarrow R_2$

$$\begin{bmatrix} 1 & 4 & 9 & : & 16 \\ 2 & 1 & 1 & : & 10 \\ 3 & 2 & 3 & : & 18 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - 3R_1$
$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -24 & -30 \end{array}\right]$$

$R_3 \to 7R_3 - 10R_2$
$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & 0 & 2 & 10 \end{array}\right]$$

$R_3 \to \frac{1}{2} R_3$
$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & 7 & -17 & 22 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$R_2 \to R_2 + (17)R_3$
$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & 7 & -0 & -63 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$\sim R_2 \to \frac{1}{7} R_2$
$$\left[\begin{array}{ccc:c} 1 & 4 & 9 & 16 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$\sim R_1 \to R_1 - 4R_2$
$$\left[\begin{array}{ccc:c} 1 & 0 & 9 & 52 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$\sim R_1 \to R_1 - 9R_3$
$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

$\boxed{x=7}, \boxed{y=-9}, \boxed{z=5}$

Solve using Gauss Jordan elimination method. (2021 PP)

$3x_1 + x_2 - x_3 = -4$

$x_1 + x_2 - 2x_3 = -4$

$-x_1 + 2x_2 - x_3 = 1$

Given System:

$x_1 + x_2 - 2x_3 = -4$ — (i)

$3x_1 + x_2 - x_3 = -4$ — (ii)

$-x_1 + 2x_2 - x_3 = 1$ — (iii)

$$\begin{bmatrix} 1 & 1 & -2 \\ 3 & 1 & -1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -4 \\ 1 \end{bmatrix}$$

$AX = B$

$$A = (A:B) = \left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 3 & 1 & -1 & -4 \\ -1 & 2 & -1 & 1 \end{array}\right]$$

$R_3 \to R_3 + R_1$

$$\left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 3 & 1 & -1 & -4 \\ 0 & 3 & -3 & -3 \end{array}\right]$$

$-\frac{1}{3} R_1 \to R_1$ $\left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 3 & 1 & -1 & -4 \\ 0 & -1 & 1 & 1 \end{array}\right]$

$R_2 \longleftrightarrow R_3$ $\left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 0 & -1 & 1 & 1 \\ 3 & 1 & -1 & -4 \end{array}\right]$

$R_3 \to R_3 - 3R_1$
$(-1)R_2 \to R_2$ $\left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & 5 & 8 \end{array}\right]$

$R_3 \to R_3 - 2R_2$ $\left[\begin{array}{ccc:c} 1 & 1 & -2 & -4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 3 & 6 \end{array}\right]$

$\frac{1}{3} R_3 \to R_3$ $\left[\begin{array}{ccc:c} 1 & 1 & -2 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$

B

$R_1 \to R_1 + R_3$ $\left[\begin{array}{ccc:c} 1 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$

$R_2 \to R_2 + R_3$

$\sim R_1 \to R_1 - R_3$

$$\left[\begin{array}{ccc:c} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 \to R_1 - R_2$

$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\boxed{x_1 = -1}$

$\boxed{x_2 = 1}$

$\boxed{x_3 = 2}$

verification:

$x_1 + x_2 - 2x_3 = -4$

$-1 + 1 - 4 = -4$

$\boxed{-4 = -4}$

(2015 PP)

$x_1 + 2x_2 - 3x_3 = 6$

$2x_1 - x_2 + 4x_3 = 1$

$x_1 - x_2 + x_3 = 3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & -1 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 3 \end{bmatrix}$$

$AX = B$

$$(A:B) = \left[\begin{array}{ccc:c} 1 & 2 & -3 & 6 \\ 2 & -1 & 4 & 1 \\ 1 & -1 & 1 & 3 \end{array}\right]$$

$R_1 \to R_1 + R_3$
$R_2 \to R_2 + R_1$

$$\left[\begin{array}{ccc:c} 1 & -1 & 1 & 3 \\ 0 & 1 & 2 & 11/5 \\ 0 & -3 & 4 & -3 \end{array}\right]$$

$R_3 \to R_3 + 3R_2$

$$\left[\begin{array}{ccc:c} 1 & -1 & 1 & 3 \\ 0 & 1 & 2 & 11/5 \\ 0 & 0 & 10 & 18/5 \end{array}\right]$$

$R_1 \to R_1 + R_2$
$\frac{1}{10} R_3 \to R_3$

$$\left[\begin{array}{ccc:c} 1 & 0 & 3 & 26/5 \\ 0 & 1 & 2 & 11/5 \\ 0 & 0 & 1 & 18/50 \end{array}\right]$$

$R_1 \to R_1 + (-3)R_3$

$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & 103/25 \\ 0 & 1 & 2 & 11/5 \\ 0 & 0 & 1 & 18/50 \end{array}\right]$$

$R_2 \to R_2 - 2R_3$

$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & 103/25 \\ 0 & 1 & 0 & 37/25 \\ 0 & 0 & 1 & 18/50 \end{array}\right]$$

$x_1 = 103/25$, $x_2 = 37/25$, $x_3 = 18/5$

<u>verification:</u>

$$x_1 - x_2 + x_3 = 3$$

$$\frac{103}{25} - \frac{37}{25} + \frac{18}{50} = 3$$

$$\frac{66}{25} + \frac{18}{50} = 3$$

$$\frac{132+18}{50} = 3 \Rightarrow \frac{150}{50} = 3$$

$$\boxed{3 = 3}$$

... that is formed by taking the coefficients of the variables and also the constant term from the system of linear equations (standard form) no of columns

Q. Define linear equation.

An equation in which the variable highest power is 1.

e.g $x+2=3$

$x+y=5$

In general linear equation:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where b and the coefficients $a_1, \dots a_n$ are real or complex numbers that are usually known in advance

Q. Define System of linear equation

=> A system (group) of two or more linear equations involving the same variable is called system of linear equation (linear system)

e.g i) $\begin{cases} x + 2y = 7 \\ x - 3y + z = 8 \end{cases}$

ii) $\begin{cases} 2x_1 + x_2 + 3x_3 = 7 \\ -4x_1 + 3x_2 + 4x_3 = 8 \\ 7x_1 - x_2 + 5x_3 = 2 \end{cases}$

=> In general:

$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$

$a_{21}x_1 + a_{22}x_2 + \dots + x_{2n}x_n = b_2$

$\vdots$

$a_{m1}x_1 + a_{mm2}x_2 + \dots + a_{mn}x_n = b_m$

So, this is general linear system of 'm' equation in 'n' variables.

Solution of linear equations:

A system of linear equation has either

=> No solution

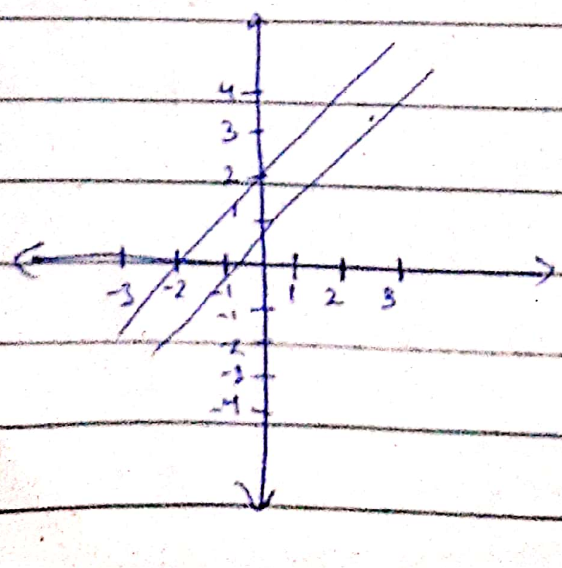
=> exactly one solution

=> Infinite many solution

1) A system of linear equation has no solution

$x_1 - 2x_2 = -1$

$-x + 2x_2 = 3$



(i) A system of linear equations has exactly one solution.

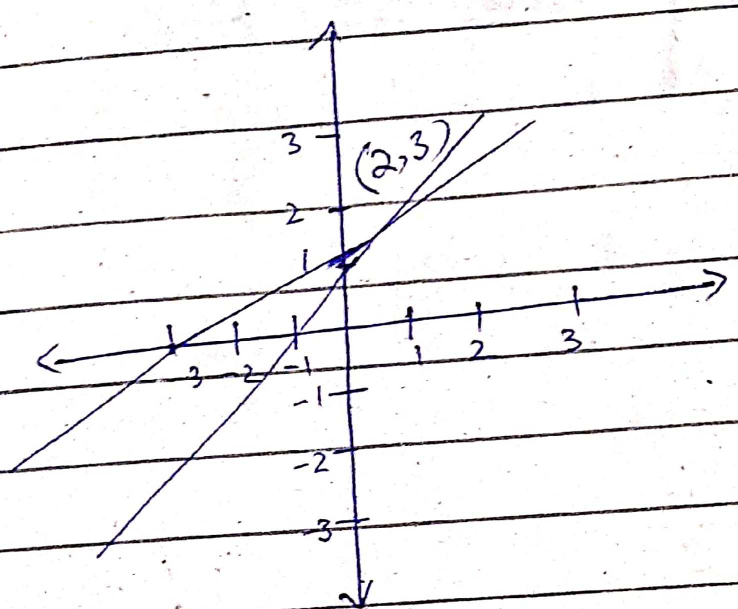
e.g

$$x_1 - 2x_2 = -1$$

$$-x_2 + 3x_2 = 3$$

$$\Rightarrow (x_1, x_2) = (2, 3)$$

$\boxed{x_1 = 2}$, $\boxed{x_2 = 3}$

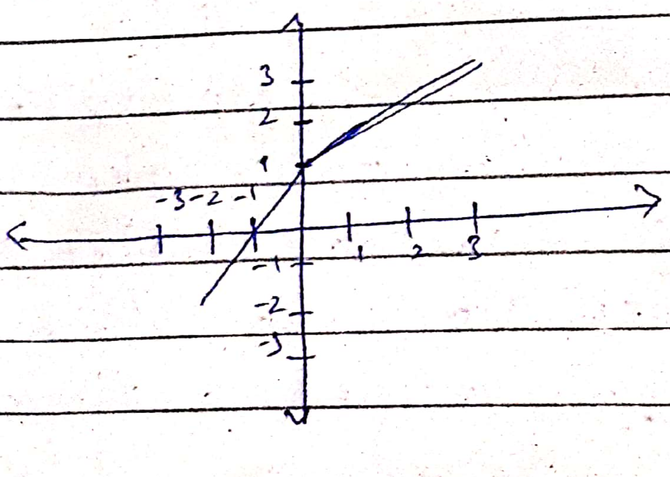


(ii) A linear system has infinitely many solutions

e.g

$$x_1 - 2x_2 = -1$$

$$-x_1 + 2x_2 = 1$$



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Note: A solution of linear system is either consistent or inconsistent.

1) Exactly one solution } → consistent
2) Infinitely many solutions }
3) No solution → inconsistence

Example: When a linear System has exactly one solution.

(Method i)

$x - y = 1$ —— (i)

$2x + y = 6$ —— (ii)

Multiplying eq (i) by (-2)

(iii) + (ii)

$-\cancel{2x} + 2y = -2$

$\cancel{2x} + y = 6$

$3y = 4$

$\boxed{y = 4/3}$

Put in (i)

$x - \frac{4}{3} = 1$

$x = 1 + \frac{4}{3}$

$\boxed{x = 7/3}$

can also be solved using row operations

Definition: A solution of the system is a list $(s_1, s_2, \ldots, s_n)$ of numbers that makes each equation a true statement when the values $s_1, \ldots, s_n$ are substituted for $x_1, \ldots$ respectively.

Example 2: When a linear system has exactly/infinitely many solution

$4x - 2y = 1$ —— (i)

$16x - 8y = 4$ —— (ii)

Multiply eq (i) by (-4)

$-16x + 8y = -4$ —— (iii)

Add (ii) & (iii)

$$\begin{array}{r} 16x - 8y = 4 \\ -16x + 8y = -4 \\ \hline \boxed{0 = 0} \end{array}$$

tell us infinitely many solutions.

Method 2:

$$Ab = \left[\begin{array}{cc|c} 4 & -2 & 1 \\ 16 & -8 & 4 \end{array}\right]$$

$-4R_1$

$$\left[\begin{array}{cc|c} -16 & 8 & -4 \\ 16 & -8 & 4 \end{array}\right]$$

$R_2 + R_1$

$$\left[\begin{array}{ccc} -16 & 8 & -4 \\ 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow \quad 0 = 0$

$\Rightarrow$ The set of all possible solutions is called solution set of the linear system.

Example 3: When a linear System has no Solution.

(Method i)

$$x+y=4 \quad \text{———} (i)$$

$$3x+3y=6 \quad \text{———} (ii)$$

Multiplying eq (i) by (-3)

$$-3x - 3y = -12 \quad \text{———} (iii)$$

Add (ii) & (iii)

$$\begin{array}{r} \cancel{3x} + \cancel{3y} = 6 \\ -\cancel{3x} - \cancel{3y} = -12 \\ \hline \boxed{0 = -6} \end{array}$$

This is not possible so system has no solution.

Method 2

$$A_b = \left[\begin{array}{cc|c} 1 & 1 & 4 \\ 3 & 3 & 6 \end{array}\right]$$

$-3R1$ $\left[\begin{array}{cc|c} -3 & -3 & -12 \\ 3 & 3 & 6 \end{array}\right]$

$R2 + R1$

$$\left[\begin{array}{cc|c} -3 & -3 & -12 \\ 0 & 0 & -6 \end{array}\right]$$

$$0 = -6$$

=> This is not possible, so, system has no solution.

Conclusion:

$$Ab = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & \cdot & 0 & 0 \end{array}\right]$$

infinitely many solution

$$Ab = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & 0 & \text{non-zero no} \end{array}\right]$$

no solution

$$Ab = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & \text{non-zero no} & \text{any no} \end{array}\right]$$

exactly one solution

Q: What is standard form of linear equation?

=> The standard form of a linear equation, also known as the "general form", is:

e.g. should be +ive $\boxed{ax} + by = c$

=> To change an equation written in slope-intercept form ($y = mx + b$) to standard form, we must get the x and y on the same side of the equal sign and a constant on the other side.

=> The standard form of a linear equation puts the x and y on the left side of the equation and makes the coefficient of the x-term

Topic:

System of homogeneous linear equation:

=> if constant term equals 0.

=> $AX = 0$ is called homogenous system where O is null matrix & $A_{m \times n}$.

=> The system $AX = B$ of m linear equation in n unknowns is known as the system of homogeneous linear equations if $B = 0$. Thus the above equation reduces to $AX = 0$.

=> For such a system matrix A and augmented matrix $[A:B]$ are the same, so the rank of coefficient matrix A and is augmented matrix are equal i.e $\rho[A:B] = \rho[A]$. Hence a system of homogeneous linear equation is always consistent.

Trivial Solution	**Non-Trivial Solution**
=> $\lvert A \rvert \neq 0$	=> $\lvert A \rvert = 0$
o if $\rho(A) =$ no of unknown, then system is consistent with unique solution or trivial solution.	o if $\rho(A) <$ no of unknowns, then system is consistent with infinite many solution. Non-trivial solution.

⇒ if $n = r$, then the eqn. $AX = 0$ will have non-linearly independent solutions ⇒ $X = 0$ ⇒ $x_1 = x_2 = x_3 = \dots = x_n = 0$ (This is the only solution) ⇒ called trivial solution.	⇒ if $n > r$, then the eqn $AX = 0$ will have $n - r$ linearly independent solutions. - called non-trivial solution. - The equation $AX = 0$ will have infinite solutions.

Example:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0$$

$$AX = 0$$

where $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$

is called coefficient matrix of order $m \times n$.

Example:

(Method 1)

$4x + 2y + z + 3u = 0$
$6x + 3y + 4z + 7u = 0$
$2x + y + u = 0$

$$[A] = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$$= R_1 \longleftrightarrow R_3 \begin{bmatrix} 2 & 1 & 0 & 1 \\ 6 & 3 & 4 & 7 \\ 4 & 2 & 1 & 3 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 3R_1$
$R_3 \rightarrow R_3 - 2R_1$

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$R_2 \rightarrow R_2/4$
$R_3 \rightarrow R_3 - R_2$

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank $(A) = 2 < 4 =$ no of Unknowns

So system is consistent and have $4-2$ independent solution

Now system become

$$2x + y + U = 0$$

$$z + U = 0$$

Let select $\boxed{z = k_1}$

$$\boxed{U = -k_1}$$

then $2x + y - k_1 = 0$

Again let $y = k_2$

$$2x + k_2 - k_1 = 0$$

$$\boxed{x = k_1 - k_2/2}$$

Method 2:

$$2x + y - z = 0 \quad \text{———} (i)$$

$$x + y - z = 0 \quad \text{———} (ii)$$

$$x + 2y + 2z = 0 \quad \text{———} (iii)$$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$|A| = 2\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$$

$$= 2(2+2) - 1(2+1) - 1(2-1)$$

$$= 8 - 3 - 1$$

$$= \cancel{4} \neq 0$$

$\therefore$ determinant of $|A| \neq 0$ so, the system has trivial solution

$\Rightarrow$ Subtraction equation (ii) and (iii)

$$\cancel{x} + y - z = 0$$
$$\underline{\pm \cancel{x} \pm 2y \mp 2z = 0}$$
$$-y - 3z = 0$$
$$\boxed{y = -3z}$$

$\Rightarrow$ Subtract eq (i) & (ii)

$$2x + y - z = 0$$
$$\underline{\pm x \pm y \mp z = 0}$$
$$\boxed{x = 0}$$

put $x = 0$ in (ii)

$$0 + (3z) - z = 0$$
$$-4z = 0 \Rightarrow \boxed{z = 0}$$

$$\boxed{y = 0}$$

$\{x, y, z\} = \{0, 0, 0\}$

(ii) $x + y + 2z = 0$ ——— (i)

$-2x + y - z = 0$ ——— (ii)

$-x + 5y + 4z = 0$ ——— (iii)

$$\begin{bmatrix} 1 & 1 & 2 \\ -2 & 1 & -1 \\ -1 & 5 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$AX = 0$

$$|A| = \begin{vmatrix} 1 & 1 & 2 \\ -2 & 1 & -1 \\ -1 & 5 & 4 \end{vmatrix}$$

$$= 1\begin{vmatrix} 1 & -1 \\ 5 & 4 \end{vmatrix} - 1\begin{vmatrix} -2 & -1 \\ -1 & 4 \end{vmatrix} + 2\begin{vmatrix} -2 & 1 \\ -1 & 5 \end{vmatrix}$$

$= 1(4+5) - 1(-8-1) + 2(-10+1)$

$= 9 + 9 - 18 = 0$

$\therefore$ $|A|$ is singular

$\Rightarrow$ So the system has non-trivial Solution.

subtract (i) & (ii)

$$\begin{array}{r} x + y + 2z = 0 \\ \underset{+}{-}2x \underset{-}{+} y \underset{+}{-} z = 0 \\ \hline 3x + 3z = 0 \end{array}$$

$3x = -3z \Rightarrow \boxed{x = -z}$

Add eq (i) & (iii)

$$\cancel{x} + y + 2z = 0$$
$$-\cancel{x} + 5y + 4z = 0$$

$$6y + 6z = 0$$
$$6y = -6z$$
$$y = -\frac{6}{6}z$$
$$\boxed{y = -z}$$

Let $z = t$

$$y = -t$$
$$\boxed{x = -t}$$